Kushagra Vyas

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.30
Shannen High School Vadodara	Class XII (CBSE)	2021	72.20%
St Xavier's High School Gurgaon Sector 49	Class X (CBSE)	2019	91.00%

PUBLICATION

Recovering Coherent-Error Information from Randomized-Compiling Twirl Labels

Posted on arXiv | DOI: 10.48550/ARXIV.2607.26756

- **Proved** an exact conservation law demonstrating that randomized compiling relocates coherent-error information.
- **Engineered** a custom state-vector simulator, validating system stability through **58** automated unit tests.
- **Recovered** target hardware parameters to within **0.0063 radacross 1,620real IBM Quantum circuit executions**.

PPG-ATM-ViT-Net: Vision Transformer for Multi-Class Arrhythmia Detection

Biomedical Signal Processing and Control, Vol. 127, Elsevier | DOI: 10.1016/j.bspc.2026.111105

- **Co-authored** a Vision Transformer framework achieving **88.62%** six-class arrhythmia accuracy.
- **Validated** the model across **91 subjects** and **46,827** PPG segments using rigorous subject-wise cross-validation.

RESEARCH EXPERIENCE

Junior Adjunct Researcher, JST LOTUS Programme

Frontier Research Institute for Interdisciplinary Sciences, Tohoku University | *Host: Prof. Le Bin Ho*

- **Developing** real-time adaptive microwave pulse-compilation optimized for qubit compilation with phase tracking.
- **Virtual-Z corrections** allow for mean iSWAP fidelity to be restored from **F 0.24** without correction to **F 0.97**.

Fujitsu Quantum Simulator Challenge 2025-26

Selected team of two representing IITH

- **Characterizing** the effects of crosstalk on neighboring qubits and designing algorithms to mitigate crosstalk.
- **Studying** whether a job can leak information about earlier computations, and developing a post-job reset protocol.

INTERNSHIPS

AI Simulation Intern, Apriori

Indian Institute of Technology Delhi Founded Startup

- **Designed** RL-driven synthetic personas, **accelerating A/B testing cycles by 68%** for partner client systems.
- **Evaluated** simulation-driven strategies, **enabling 26% faster transitions** from production to deployment.

AI Hardware Security Research Intern

PUREID Security Lab (Acquired by ColorTokens)

- **Architected and evaluated** an AI-driven hardware pipeline, **reducing authentication overhead by 17%**.

- **Enhanced** authentication robustness by up to **30%** through the training and deployment of an in-house model.

RESEARCH PROJECTS

DALT: Reinforcement-Learning-Based Adaptive Quantum Transpiler

M.Tech. research, IIT Hyderabad |Link: github.com/kushcoder12/DALT | *Advisor: Prof. Sangkha Borah*

- **Trained a Double-DQN agent** to adapt transpilation decisions under simulated hardware drift.

- **Tested** cross-job parameter-drift persistence in three campaigns **(54 jobs)** on **IBM's 156-qubit Heron processor.**

POSITIONS OF RESPONSIBILITY

Teaching Assistant, Quantum and Solid State Devices Laboratory

Computational Lab | IIT Hyderabad

- **Teaching** physics simulation, Python, and Qiskit-based quantum programming to M.Tech. lab cohorts.

- **Design and supervise** hands-on experimental simulations, VQE implementations, and quantum error mitigation.

CONFERENCE PRESENTATIONS

Real-Time Adaptive Pulse Compilation for Drifting Quantum Hardware

Poster Presentation at Quantum Innovation 2026, Nagoya, Japan

Developed Kalman-filter-driven pulse framework with phase tracking, across 7 simulated chip topologies (3-20 qubits).

TECHNICAL SKILLS

Quantum: Qiskit, IBM Quantum hardware, pulse-level control, randomized compiling, noise modeling

Methods: Fisher information, Kalman filtering, reinforcement learning, hypothesis testing

Programming: Python, NumPy, PyTorch, JAX, Git, Linux, L A T E X

AWARDS AND HONORS

- JST LOTUS research fellowship, Japan Science and Technology Agency, 2026.

- Undergraduate Research Award (URA-1) for PPG-ATM-ViT-Net, IITRAM, 2024.